

Mid-Research Report

S&P 500 Factor Backtest — 5 Libraries $\times$ 2 Cost Regimes, 2016–2025

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14 September 2026

Test window: 2016-01-01 .. 2025-12-26. **Benchmark:** cap-weighted S&P 500 (GSPC), total-return basis. **Data:** `us_data` (Qlib). **5 libraries** under `data/factorlib/`. Every return, Sharpe, Calmar and cumulative-excess figure below is an *excess* return over the S&P 500 benchmark (total-return basis unless a table states otherwise); no raw portfolio return is reported as if it were excess.

BEST SHARPE	BEST 10Y EXCESS	BEST RANK IC	FRESH-ALPHA @1Q	CELLS LIVE
−0.02	−4.82%	0.0094	−2.18%	10 / 10
US-MAIN · FLAT	US-MAIN · FLAT	0.58 $\times$ topk hurdle	first quarter after deploy	5 FLAT · 5 FULL

Headline tiles. Best Sharpe and best 10-year *excess* are the walk-forward US-MAIN FLAT book (§1.3); the −20.59% this report previously carried was a frozen-train artifact. Best Rank IC is the frozen US-MAIN FULL cell. Fresh-alpha @1Q is the frozen US-MAIN FULL decay (§11). All 10 planned cells are now live. The six CSI-mined cells all lose on the US total-return hurdle (best CSI excess CSI-ZOO FLAT −9.48%, still below break-even); the edge is local to CSI300, consistent with the prior transfer test (§12).

1 Setup

This system uses a large language model to invent stock-picking formulas. It was built and tuned on China’s CSI300; the question here is whether the formulas it mines beat the S&P 500 once trading is charged realistically. Five libraries are tested, each under two cost regimes, over the ten years 2016–2025. All returns are excess over the S&P 500 total-return benchmark.

1.1 The five libraries

Tag	From	Count	What it is
US-MAIN	US mine (deepseek)	150	every formula the US deepseek run generated
US-ZOO	US mine (deepseek)	8	the subset its quality gate admitted — what it would trade
CSI-MAIN	CSI300 mine (main)	150	every formula the CSI300 main run generated
CSI-ZOO	CSI300 mine (main)	56	the subset its quality gate admitted
CSI-ORIG	CSI300 paper-orig.	150	the original paper-baseline library

Table 1: The five collections compared throughout. US-MAIN/US-ZOO come from the `meanvar_20260911_223440` US run; CSI-MAIN/CSI-ZOO from the `meanvar_20260828_194432` CSI300 run; CSI-ORIG from `all_factors_library_original_20260831_012324`. **us-main vs us-zoo tests whether the quality gate earns its keep; us-main vs csi-main on US data tests method vs market (§12).**

1.2 Two harnesses, one construction family

Both regimes use a LightGBM combiner feeding a top-50 long-only portfolio, but the two harnesses differ in what they can charge and in their label — a confound flagged in §15.

FLAT cost — `btv2`.

The Qlib backtest harness applies a flat fee through `exchange_kwargs` (open/close cost, no slippage, no impact). It trains the combiner on a close-to-close label $\text{Ref}(\$close, -2)/\text{Ref}(\$close, -1) - 1$, with a 2-step walk-forward train (§1.3): 2010–2014 (valid 2015) for the 2016–2021 test slice, retrained on 2016–2020 (valid 2021) for the 2022–2025 slice.

FULL cost — `eval operator`.

The evaluation operator applies the full kappa slippage+impact model, trains the combiner on 2005–2012, validates on 2013–2015, tests on 2016–2025, on an open-to-open label $\text{Ref}(\$open, -2)/\text{Ref}(\$open, -1) - 1$ filled at next open. The benchmark is the S&P 500 on a total-return basis *natively* (`benchmark_basis: estimated_total`).

Remark 1 (Neither system saw the test period). *The mining valid window ends 2015-12-31; the test window 2016-01-04 onward is strictly out of sample. The FLAT harness’s two train windows (2010–2014 and 2016–2020) and the FULL harness’s train window (2005–2012) all end before their respective test slices. No result below is contaminated by hindsight.*

1.3 Train window: frozen vs walk-forward

The FLAT (`btv2`) numbers in this report use a **2-step walk-forward** combiner on the *same* factor library `meanvar_20260911_223440` (150 factors, and its 8-factor `..._zoo` admitted subset):

- 2016–2021: the combiner trained on 2010–2014 (valid 2015) — still fresh at the 2016 deployment, decaying through 2021.
- 2022–2025: the combiner retrained on 2016–2020 (valid 2021) — fresh at the 2022 deployment.

Both slices use the same LightGBM combiner, the same top-50 long-only `TopkDropoutStrategy` (topk 50, n_{drop} 5), and the same flat ~ 1 bp cost. The two slices are stitched on the report’s total-return basis (`btv2`’s native GSPC benchmark is price-only, so the S&P 500 dividend yield, 1.807%/yr, is subtracted from the daily excess). The 2022–2025 slice reproduces the stand-alone +9.03% *price-bench* result exactly (Rank IC 0.0112, IC 0.0115), confirming the re-run is correct; on the total-return basis that slice is +1.83%.

Why walk-forward. The original 2016–2025 `btv2` run trained the combiner *once* on 2010–2014 and froze it for the whole test — so by 2022–2025 the model was 8–12 years stale, and the per-year excess deteriorated (+12.05/−13.07/−4.83/−8.62%, total −15.14% for 2022–2025). A fresh 2016–2020 combiner on the same factors turns that into +7.01/−9.58/+0.37/+4.84% (total +1.83% excess, price-bench +9.03%) — a 15.8pp swing from re-fitting the combiner alone, with no new factors. The frozen-train total of −20.59% was therefore a staleness artifact; the walk-forward total is −4.82% (break-even). The 8-factor zoo shows the same effect (frozen −28.48% → walk-forward −22.19%).

The FULL regime is still frozen. The eval-operator (FULL) combiner trains once on 2005–2012 and is frozen through 2025 — the same staleness, more severe (10–20 years stale by 2022–2025). Its -46.40% is therefore likely overstated by the same mechanism. A walk-forward FULL re-run (~ 65 min per library) is deferred; the FLAT-vs-FULL comparisons below flag the resulting train-window confound rather than present a clean cost isolation.

1.4 Benchmark: total return, not price

The long-only top-50 book reinvests its dividends, so it is a *total-return* stream. The benchmark must therefore be the S&P 500 *total* return, not the price index — otherwise the book is credited the market’s $\sim 1.8\%/yr$ dividend yield as fake alpha. The FULL harness handles this natively (`estimated_total`); the FLAT harness benchmarks against GSPC price return, so the report subtracts the estimated S&P 500 dividend yield (mean $1.81\%/yr$ over the window) from the FLAT price-bench excess in post. Every excess figure in the headline and per-year tables is on this total-return basis; the one exception is the raw price-bench column in §8, which is explicitly labelled.

2 Cost Models

The US market is *spread- and impact-dominated*: zero retail commission, no stamp duty. The cost model therefore lives in the slippage and impact terms, not in a flat fee. Every constant below is calibrated to a measured 2024 US S&P 500 figure, not fitted to this backtest.

2.1 FLAT cost (btv2, Qlib exchange_kwargs)

Constant	Value	Rationale
<code>open_cost</code>	0.00005 (0.5 bp)	One-way buy commission. US has no stamp duty; zero-commission retail brokers charge ~ 0 . Institutional all-in (commission + SEC Section 31 fee $\$27.80/\$1M \approx 0.278$ bp sell-side + FINRA TAF) is ~ 0.3 – 0.5 bp. 0.5 bp is a conservative institutional one-way estimate.
<code>close_cost</code>	0.00005 (0.5 bp)	Symmetric sell-side commission (same rationale as open). The US has no stamp duty on the seller either, so sell = buy.
<code>min_cost</code>	0	Minimum ticket charge per trade. US brokers have no per-ticket minimum under zero-commission.
<code>limit_threshold</code>	1.0	Daily price-limit threshold for execution. The US has no daily price-limit board (a name can move $> 10\%/day$ on earnings), so 1.0 = no limit.
<code>deal_price</code>	open	Fill at next open, matching the eval operator’s <code>fill_rule: open_next</code> .
<code>benchmark</code>	GSPC	Cap-weighted S&P 500 (Qlib instrument GSPC); price return, converted to total-return in post (§1).

Table 2: FLAT-cost constants. **Round-trip** = $2 \times 0.5 = 1$ **bp**, the US zero-commission / zero-stamp floor (commission only).

2.2 FULL cost (eval operator, kappa slippage + impact)

The full cost model is

$$\text{charges} = \kappa_0 |\Delta w| + \kappa_1 \sum \sigma |\Delta w| + \kappa_2 \varphi, \quad \varphi \propto \frac{|\Delta w|^{1.5}}{\sqrt{\text{ADV}}},$$

adding slippage (half-spread) and market impact on top of the flat commission — the terms that dominate US trading costs.

Constant	Value	Rationale
κ_0	0.00005 (0.5 bp)	Flat per-one-way-turnover commission (same as <code>btv2</code>). Institutional commission + SEC fee + FINRA TAF $\approx$ 0.3–0.5 bp; $\sim$ 0 under zero-commission retail. Kept at the conservative institutional level.
κ_1	0.012	Slippage = $\kappa_1 \sum \sigma \Delta w $. At median US large-cap $\sigma \approx$ 1.3%/day this gives $\sim$ 1.6 bp one-way, inside the 1–3 bp half-spread band (QuanterLab).
κ_2	0.003	Impact $\kappa_2 \sqrt{p}$ at 1% participation $\rightarrow$ 3 bp, inside the 2–5 bp @1% ADV band (Kissell 2024 Q1: $\sim$ 17 bp total large-cap cost at 10% ADV).
impact exponent	1.5	$\varphi \propto \Delta w ^{3/2} / \sqrt{\text{ADV}}$ — the square-root impact law (Almgren–Chriss / Kissell).
β_{day}	0.0004	$\sim$ 10%/yr borrow cost. Inert while <code>signed:false</code> (long-only — no shorts, no borrow charge).
<code>nav</code>	10^8 (\$100M)	Book NAV; normalises ADV into a participation rate (the impact term is scale-invariant under this convention).
<code>adv_scale</code>	1.0	<code>dollar_volume = panel.amount $\times$ adv_scale</code> . USD: <code>us_data \$amount</code> is already true USD notional, so scale = 1.0.
<code>min_cap</code>	10^8 (\$100M)	ADV floor below which a name is too illiquid to trade.
<code>benchmark / basis</code>	GSPC / <code>estimated_total</code>	S&P 500 on a total-return basis: price return + estimated dividend yield.

Table 3: FULL-cost constants, calibrated to measured 2024 US S&P 500 figures. **Round-trip** $\approx 2 \times (0.5 + 1.6 + 3) \approx 10$ **bp** at 1% ADV, matching the 5–10 bp retail / $\leq$ 20 bp institutional round-trip targets (Kissell 2024 Q1, QuanterLab, JoF 2025 “Actual Retail Price”).

3 What the Metrics Mean

A *factor* scores every stock each day; the *portfolio* buys the 50 highest-scoring ones, equally weighted. Two questions matter: does the score predict what happens next, and does trading on it make money after costs? Every return metric below is an excess return over the S&P 500 total-return benchmark.

4 Headline Results

All numbers are computed on 2016-01-01 – 2025-12-26, out of sample (§1), excess over the S&P 500 total-return benchmark. The combined table below mirrors the HTML report: one row per library $\times$ regime (10 cells = 5 libraries $\times$ 2 cost regimes), all return and prediction metrics in one place.

Metric	What it means	Range	Good is
Sharpe	Annualised Sharpe of the daily <i>excess</i> return.	unbounded	> 1 ; > 2 strong
ARR	Annualised <i>excess</i> return (geometric).	unbounded	$> 5\%$
CRR (Calmar)	ARR divided by $ \text{MDD} $.	unbounded	> 0.5
IC	Cross-sectional correlation of score with next-return, per day.	-1 to 1 ; real 0.01 – 0.10	> 0.03
ICIR	IC mean / IC std — consistency of the edge day to day.	usually 0 – 1	> 0.3
Rank IC	Same as IC using only the <i>order</i> of scores; the number that matters for a top-50 book.	-1 to 1	> 0.03
Rank ICIR	Rank IC mean / std.	usually 0 – 1	> 0.3
MDD	Worst peak-to-trough loss along the way (of the excess curve).	-100% to 0	better than -20%
Turnover	Share of the portfolio traded daily — the multiplier on every cost.	0 to 1	$< 0.20/\text{day}$
Cost (bp)	Trading friction, basis points traded.	≥ 0	< 10
cum. excess	Cumulative <i>excess</i> return over the period (not annualised).	unbounded	> 0
total excess	Cumulative excess return over the 10y window vs S&P 500 total return.	unbounded	> 0

Table 4: Every measure used in this report. “Good is” are conventional rules of thumb, not thresholds the system enforces.

Why n differs between the FLAT and FULL columns. The two regimes run through different backtest harnesses that count factors at different granularities, so the n column is not expected to match cell-to-cell. *FLAT* (`btv2`, `LightGBM+topk`) loads every library entry *by name* — duplicate formulas become separate (identical) feature columns, which is harmless for the tree model. *FULL* (the eval operator) deduplicates by expression string before loading, so identical formulas collapse to one signal, and any factor that parses but errors at recompute is dropped. The per-library gap:

- US-MAIN: 150 entries, 145 *unique* formulas — 5 duplicate copies of 3 formulas (252-day market beta mined under 4 names, 21-day idiosyncratic skewness under 2, capital-gains-overhang under 2). FLAT $n=150$, FULL $n=145$: gap is dedup.
- US-ZOO: 8 unique, no duplicates. $n=8$ both.
- CSI-MAIN: 150 entries, 147 unique (3 duplicate formulas). FLAT $n=150$, FULL $n=147$: gap is dedup.
- CSI-ZOO: 56 unique, no duplicates. $n=56$ both.
- CSI-ORIG: 150 entries, 150 unique, but 3 factors use bare cross-sectional operators (`STD`, `MEDIAN` — distinct from the time-series `TS.STD`) that the eval recompute path, which evaluates per-

Lib · regime	<i>n</i>	Sharpe	ARR%	CRR	IC	ICIR	RankIC	RankICIR	MDD%	turn	cost	total%
US-MAIN · FLAT	150	−0.02	−0.49	−0.02	0.0095	0.055	0.0093	0.054	−20.55	—	—	−4.82
US-MAIN · FULL	145	−0.24	−6.07	−0.12	0.0053	0.046	0.0094	0.080	−49.05	0.104	1.16	−46.40
US-ZOO · FLAT	8	−0.33	−2.49	−0.09	0.0092	0.045	0.0090	0.045	−28.48	—	—	−22.19
US-ZOO · FULL	8	−0.24	−5.95	−0.12	0.0050	0.043	0.0087	0.076	−48.75	0.105	1.21	−45.74
CSI-MAIN · FLAT	150	−0.31	−3.12	−0.09	0.0073	0.040	0.0069	0.038	−35.52	—	—	−27.05
CSI-MAIN · FULL	147	−0.19	−4.96	−0.11	0.0062	0.058	0.0086	0.082	−45.78	0.104	1.16	−39.79
CSI-ZOO · FLAT	56	−0.06	−1.00	−0.03	0.0073	0.048	0.0069	0.046	−32.77	—	—	−9.48
CSI-ZOO · FULL	56	−0.31	−6.86	−0.13	0.0047	0.045	0.0072	0.070	−54.46	0.104	1.10	−50.74
CSI-ORIG · FLAT	147	−0.19	−2.13	−0.08	0.0060	0.039	0.0061	0.040	−27.88	—	—	−19.30
CSI-ORIG · FULL	147	−0.21	−5.36	−0.12	0.0037	0.036	0.0046	0.044	−46.39	0.104	1.18	−42.24

Table 5: Return, risk and prediction quality — total over 2016–2025, excess over the S&P 500 total return. ARR/Sharpe/CRR/MDD/total are excess-return quantities; IC/ICIR/RankIC/RankICIR are cross-sectional. FLAT turnover is not separately reported by *btv2* (its flat fee is 1 bp round-trip, “rt”); FULL turnover $\approx 0.104/\text{day}$ and cost $\approx 1.2\text{ bp}$. The least-bad cell is US-MAIN FLAT (−4.82%, essentially break-even, Sharpe −0.02); the worst is CSI-ZOO FULL (−50.74%). All six CSI-mined cells lose on the US total-return hurdle (excess −9.48% to −50.74%), confirming the edge is local to CSI300 (§12). **Conclusion: no live cell is positive on the total-return basis — but the walk-forward us-main FLAT book beats the S&P 500 *price* index (+14.74%, Sharpe +0.21, not significant) and is break-even on total return.**

instrument, cannot compute (`KeyError: 'instrument'`). FLAT $n=150$, FULL $n=147$: gap is load-failure.

So the n discrepancy is deduplication (and, for CSI-ORIG, three compute-failures), not missing data: both harnesses see the same formulas, counted per-entry (FLAT) vs per-unique-formula (FULL). The per-year headers disclose each cell’s count (“150 factors” / “145 factors loaded” / “147 of 150 factors loaded”).

5 Per-Year Results

Each table is one library; within it the FLAT and FULL blocks are separated by a `\midrule`. Columns are as in the HTML report, including the final **cum. excess** column (the period’s actual cumulative excess return, not annualised; for a full year this is close to ARR, for the partial 2025 it differs). All ARR/Sharpe/CRR/MDD/cum. excess are on the daily *excess* return.

6 Forward Test: 2022–2025 (Frozen Train pre-2022)

The 2016–2025 results above use a 2-step walk-forward combiner (US libraries) or a frozen train ending in 2014 (CSI libraries). A stricter recency question is whether *any* library clears the U.S. hurdle when the combiner is trained once on all pre-2022 data and tested on the most recent four years. We re-run all five libraries through the *btv2* FLAT harness (LightGBM + TopkDropout-Strategy, 1 bp round-trip flat fee, close-to-close label; the GSPC price benchmark has the S&P 500 dividend yield subtracted to a total-return basis, as in the main tables) with a single frozen split:

- **train** 2009-01-01–2018-12-31 (10y; the factor cache begins 2008-12-29, so 2009 is the first full year of signal),

year	ARR%	Sharpe	CRR	IC	ICIR	Rank IC	Rank ICIR	MDD%	cum. exc.%
US-MAIN (150 factors) — FLAT cost (btv2), 2-step walk-forward									
2016	−0.40	−0.02	−0.06	0.0011	0.005	0.0028	0.013	−6.25	−0.40
2017	2.13	0.46	0.41	0.0118	0.094	0.0117	0.095	−5.26	+2.12
2018	0.48	0.11	0.11	0.0071	0.060	0.0077	0.064	−4.27	+0.48
2019	0.94	0.16	0.13	0.0085	0.042	0.0077	0.038	−7.32	+0.94
2020	−7.42	−0.52	−0.67	0.0205	0.088	0.0186	0.080	−11.12	−7.44
2021	−2.06	−0.17	−0.18	−0.0003	−0.002	0.0002	0.001	−11.13	−2.06
2022	7.01	0.89	1.37	−0.0061	−0.039	−0.0072	−0.045	−5.13	+6.98
2023	−9.58	−1.18	−0.84	0.0085	0.054	0.0085	0.054	−11.37	−9.50
2024	0.37	0.09	0.05	0.0173	0.133	0.0167	0.129	−7.70	+0.37
2025	4.84	0.59	0.95	0.0267	0.170	0.0270	0.170	−5.11	+4.74
TOTAL	−0.49	−0.02	−0.02	0.0095	0.055	0.0093	0.054	−20.55	−4.82
US-MAIN (145 factors loaded) — FULL cost (eval operator), frozen train 2005–2012									
2016	−8.95	−0.61	−0.84	−0.0022	−0.017	0.0006	0.004	−10.68	−8.95
2017	−4.55	−0.59	−0.77	0.0083	0.084	0.0151	0.146	−5.95	−4.54
2018	−6.38	−0.46	−0.66	0.0090	0.108	0.0140	0.160	−9.74	−6.35
2019	0.21	0.07	0.03	0.0046	0.039	0.0074	0.058	−6.55	+0.21
2020	−11.04	−0.08	−0.57	0.0236	0.147	0.0297	0.196	−19.29	−11.08
2021	−6.23	−0.35	−0.59	−0.0040	−0.030	−0.0006	−0.005	−10.63	−6.23
2022	7.27	0.44	0.99	0.0020	0.018	0.0030	0.028	−7.32	+7.24
2023	−12.26	−0.91	−0.73	0.0075	0.079	0.0099	0.105	−16.87	−12.17
2024	−10.31	−0.74	−0.73	0.0051	0.052	0.0105	0.113	−14.16	−10.31
2025	−6.87	−0.35	−0.37	−0.0009	−0.008	0.0037	0.033	−18.70	−6.73
TOTAL	−6.07	−0.24	−0.12	0.0053	0.046	0.0094	0.080	−49.05	−46.40

Table 6: US-MAIN (US-mined main, deepseek), per year and total, both cost regimes. The FLAT block is the 2-step walk-forward retrain (§1.3): 2016–2021 on the frozen 2010–2014 train (still fresh at 2016), 2022–2025 on a fresh 2016–2020 train. **Conclusion: the walk-forward FLAT book is positive in six of ten years (2017–2019, 2022, 2024, 2025) and ends essentially break-even (−4.82%, Sharpe −0.02); the FULL block (still frozen-train 2005–2012, walk-forward re-run deferred) is positive in only two years and ends −46.40%.**

- **valid** 2019-01-01–2021-12-31 (3y; spans pre-COVID, COVID and the recovery, a robust early-stop window),
- **test** 2022-01-01–2025-12-26 (4y).

This is the LightGBM regime of the main results, applied uniformly to all five libraries on the same recent test window, so the comparison isolates the effect of a longer, frozen train and a recent-only test.

No library clears the U.S. hurdle on the recent four years: every total-excess cell is negative, from CSI-ZOO’s −5.17% (best) to CSI-ORIG’s −27.87% (worst). The ranking inverts relative to the 2016–2025 walk-forward headline, where US-MAIN FLAT was the best cell (−4.82%); on the 2022–2025 frozen train US-MAIN falls to fourth (−18.30%), while the two CSI libraries — CSI-ZOO (−5.17%) and CSI-MAIN (−6.55%) — are the most robust. The cross-sectional IC is positive for every library (0.008–0.015, at or above the 2016–2025 FLAT band of 0.006–0.0095), yet none converts to a positive return — the same IC–return decoupling documented in the main results, and still ~5–9× below the 0.0728 break-even IC. One caveat: the US libraries switch from walk-forward (2016–2025) to a single frozen train here, so part of US-MAIN’s drop is the loss of regime-adaptive retraining rather than pure factor decay; the CSI libraries were frozen-train in both windows, so

year	ARR%	Sharpe	CRR	IC	ICIR	Rank IC	Rank ICIR	MDD%	cum. exc.%
US-ZOO (8 factors) — FLAT cost (btv2), 2-step walk-forward									
2016	-7.52	-1.21	-0.81	-0.0151	-0.065	-0.0140	-0.061	-9.27	-7.52
2017	2.19	0.60	0.67	0.0170	0.102	0.0170	0.109	-3.27	+2.18
2018	0.08	0.04	0.02	0.0079	0.050	0.0080	0.052	-3.82	+0.08
2019	-4.59	-0.79	-0.59	0.0027	0.012	0.0024	0.011	-7.84	-4.59
2020	-1.21	-0.14	-0.20	0.0270	0.085	0.0237	0.080	-6.08	-1.21
2021	-5.50	-0.69	-0.65	-0.0014	-0.009	0.0011	0.007	-8.43	-5.50
2022	-7.00	-0.83	-0.76	-0.0088	-0.050	-0.0118	-0.066	-9.26	-6.97
2023	-1.32	-0.15	-0.20	0.0142	0.076	0.0148	0.081	-6.63	-1.31
2024	4.58	0.58	0.77	0.0251	0.135	0.0248	0.132	-5.93	+4.58
2025	-3.87	-0.42	-0.46	0.0237	0.125	0.0242	0.127	-8.38	-3.80
TOTAL	-2.49	-0.33	-0.09	0.0092	0.045	0.0090	0.045	-28.48	-22.19
US-ZOO (8 factors) — FULL cost (eval operator), frozen train 2005–2012									
2016	-1.40	-0.05	-0.30	0.0011	0.009	0.0012	0.009	-4.63	-1.40
2017	-3.47	-0.45	-0.65	0.0073	0.076	0.0117	0.114	-5.33	-3.46
2018	-7.17	-0.53	-0.70	0.0016	0.018	0.0053	0.058	-10.21	-7.15
2019	-3.66	-0.27	-0.41	0.0019	0.018	0.0053	0.047	-8.87	-3.66
2020	-13.73	-0.15	-0.69	0.0247	0.155	0.0281	0.184	-19.82	-13.78
2021	-3.42	-0.16	-0.35	-0.0004	-0.004	0.0019	0.017	-9.76	-3.42
2022	5.99	0.39	0.76	0.0021	0.016	0.0047	0.036	-7.89	+5.97
2023	-14.47	-1.09	-0.83	0.0052	0.049	0.0068	0.065	-17.41	-14.37
2024	-10.64	-0.72	-0.69	0.0039	0.042	0.0136	0.154	-15.48	-10.64
2025	-5.73	-0.28	-0.37	0.0021	0.019	0.0081	0.074	-15.65	-5.62
TOTAL	-5.95	-0.24	-0.12	0.0050	0.043	0.0087	0.076	-48.75	-45.74

Table 7: US-ZOO (the 8-factor admitted zoo), per year and total, both cost regimes. The FLAT block is the 2-step walk-forward retrain (§1.3). **Conclusion: under walk-forward FLAT cost the 8-factor zoo is materially worse than the 150-factor full library (Sharpe -0.33 vs -0.02 ; total excess -22.19% vs -4.82% ; price-bench -6.20% vs $+14.74\%$) — the extra factors do help when costs are low and training is fresh. Under FULL cost (still frozen-train) the two are indistinguishable (-45.74% vs -46.40%), so the dilution finding is regime-dependent (§10).**

their relative stability and the CSI>US ordering on 2022–2025 is not a method artifact.

The loss is concentrated in 2023 and 2025. 2022 was positive for four of five libraries (CSI-MAIN $+9.73\%$, CSI-ZOO $+8.94\%$, US-MAIN $+4.85\%$, CSI-ORIG $+2.72\%$; only US-ZOO negative at -0.97%); 2023 was uniformly poor (-8% to -15%); 2024 was roughly flat (only CSI-MAIN clearly positive at $+1.00\%$); and 2025 was negative for all five (-3% to -12%). CSI-ORIG, the CSI paper-original library and already the weakest in 2016–2025, loses the most (-27.87% total, -33.76% drawdown). The pattern mirrors the 2016–2025 per-year tables: positive cross-sectional IC in every year, negative excess return, and no year in which a majority of libraries jointly beat the benchmark. The forward test therefore corroborates the main verdict — no live U.S. edge for any library, and the US mine’s 2016–2025 “win” does not survive a clean recent-period test.

year	ARR%	Sharpe	CRR	IC	ICIR	Rank IC	Rank ICIR	MDD%	cum. exc.%
CSI-MAIN (150 factors) — FLAT cost (btv2), frozen train 2010–2014									
2016	−2.86	−0.47	−0.50	0.0030	0.015	0.0035	0.018	−5.66	−2.86
2017	3.69	0.83	1.41	0.0119	0.094	0.0106	0.085	−2.62	+3.68
2018	−1.46	−0.25	−0.18	−0.0012	−0.010	−0.0015	−0.013	−7.92	−1.45
2019	2.57	0.45	0.48	0.0030	0.015	0.0023	0.011	−5.31	+2.57
2020	−4.03	−0.26	−0.24	0.0199	0.085	0.0193	0.083	−17.13	−4.05
2021	−6.74	−0.64	−0.50	0.0042	0.022	0.0053	0.028	−13.44	−6.74
2022	8.00	0.87	1.36	0.0195	0.092	0.0179	0.086	−5.88	+7.97
2023	−11.82	−1.40	−0.85	−0.0033	−0.018	−0.0038	−0.021	−13.90	−11.73
2024	−6.16	−0.60	−0.58	0.0115	0.088	0.0117	0.089	−10.62	−6.16
2025	−10.65	−0.88	−0.55	0.0037	0.020	0.0031	0.017	−19.30	−10.45
TOTAL	−3.12	−0.31	−0.09	0.0073	0.040	0.0069	0.038	−35.52	−27.05
CSI-MAIN (147 of 150 factors loaded) — FULL cost (eval operator), frozen train 2005–2012									
2016	0.25	0.09	0.04	0.0128	0.127	0.0189	0.184	−6.22	+0.25
2017	0.09	0.05	0.02	0.0091	0.113	0.0128	0.162	−4.29	+0.09
2018	−9.86	−0.77	−0.93	0.0044	0.052	0.0070	0.081	−10.61	−9.83
2019	−1.42	−0.08	−0.15	−0.0002	−0.002	0.0002	0.002	−9.35	−1.42
2020	−9.22	−0.03	−0.39	0.0217	0.144	0.0224	0.155	−23.34	−9.25
2021	−6.08	−0.39	−0.45	0.0007	0.006	0.0021	0.020	−13.55	−6.08
2022	10.77	0.61	1.62	0.0065	0.062	0.0076	0.076	−6.63	+10.72
2023	−16.44	−1.29	−0.86	0.0062	0.060	0.0042	0.042	−19.03	−16.33
2024	−10.27	−0.76	−0.81	0.0014	0.014	0.0054	0.055	−12.63	−10.27
2025	−4.82	−0.23	−0.28	−0.0011	−0.009	0.0051	0.043	−17.14	−4.73
TOTAL	−4.96	−0.19	−0.11	0.0062	0.058	0.0086	0.082	−45.78	−39.79

Table 8: CSI-MAIN (CSI-mined main), per year and total. Both regimes lose on the US total-return hurdle: FLAT −27.05% (frozen train 2010–2014), FULL −39.79% (frozen train 2005–2012). The CSI-mined main library fails the US hurdle either way (§12).

7 FLAT vs FULL: the Cost Isolation

8 Benchmark Basis at the Cost Floor

The FLAT regime charges only 1 bp round-trip — effectively the zero-cost floor. Under the walk-forward retrain (§1.3) the US-MAIN book ends −4.82% below the S&P 500 *total* return — essentially break-even (Sharpe −0.02). The reason it is negative at all is the benchmark basis, not the cost model. Measured against the S&P 500 *price* index (before any dividend adjustment) the FLAT book is *positive*:

9 Prediction Quality vs the Break-Even

The deeper root cause is that the prediction score never clears the cost break-even. Two break-even hurdles are measured for this construction family (LightGBM combiner + top-50 long-only):

- **Rank IC** 0.0162 — the LightGBM+topk break-even, the score a library must reach to cover trading costs at this turnover.
- **IC** 0.0728 — the oracle-ladder break-even (the IC an oracle would need to clear costs without the combiner’s noise).

year	ARR%	Sharpe	CRR	IC	ICIR	Rank IC	Rank ICIR	MDD%	cum. exc.%
CSI-ZOO (56 factors) — FLAT cost (btv2), frozen train 2010–2014									
2016	3.09	0.66	0.90	0.0110	0.070	0.0114	0.072	−3.44	+3.09
2017	2.00	0.49	0.49	0.0074	0.065	0.0056	0.048	−4.06	+1.99
2018	−3.96	−0.70	−0.48	−0.0001	−0.001	−0.0007	−0.006	−8.34	−3.94
2019	−2.31	−0.37	−0.33	−0.0014	−0.009	−0.0019	−0.013	−6.95	−2.31
2020	−9.02	−0.43	−0.36	0.0133	0.068	0.0140	0.071	−25.33	−9.05
2021	3.00	0.39	0.38	0.0107	0.066	0.0097	0.060	−7.92	+3.00
2022	20.12	2.07	4.60	0.0186	0.118	0.0176	0.112	−4.38	+20.03
2023	−11.81	−1.42	−0.87	−0.0041	−0.027	−0.0045	−0.030	−13.62	−11.72
2024	−4.00	−0.35	−0.42	0.0106	0.082	0.0114	0.089	−9.54	−4.00
2025	−3.81	−0.30	−0.27	0.0070	0.046	0.0069	0.045	−14.18	−3.74
TOTAL	−1.00	−0.06	−0.03	0.0073	0.048	0.0069	0.046	−32.77	−9.48
CSI-ZOO (56 factors) — FULL cost (eval operator), frozen train 2005–2012									
2016	−2.31	−0.12	−0.34	0.0032	0.030	0.0068	0.063	−6.89	−2.31
2017	1.92	0.32	0.68	0.0063	0.074	0.0110	0.129	−2.83	+1.91
2018	−5.56	−0.40	−0.71	0.0062	0.074	0.0100	0.114	−7.82	−5.54
2019	−7.24	−0.62	−0.65	−0.0011	−0.012	−0.0018	−0.019	−11.08	−7.24
2020	−13.71	−0.16	−0.67	0.0190	0.134	0.0218	0.161	−20.44	−13.76
2021	−7.91	−0.64	−0.67	0.0024	0.020	0.0058	0.052	−11.76	−7.91
2022	1.48	0.17	0.18	0.0043	0.042	0.0042	0.042	−8.29	+1.47
2023	−14.47	−1.20	−0.81	0.0029	0.029	0.0024	0.023	−17.83	−14.36
2024	−8.25	−0.63	−0.67	0.0022	0.024	0.0069	0.071	−12.30	−8.25
2025	−11.03	−0.66	−0.58	0.0010	0.009	0.0048	0.045	−19.13	−10.83
TOTAL	−6.86	−0.31	−0.13	0.0047	0.045	0.0072	0.070	−54.46	−50.74

Table 9: CSI-ZOO (CSI-mined main zoo), per year and total. FLAT −9.48% is the best CSI cell yet still below break-even; FULL −50.74% is the worst cell overall — the 56-factor admitted zoo is worse than the 147-factor full library under FULL cost, mirroring the US US-ZOO finding (§12).

Every live cell sits below the topk break-even: the best Rank IC is 0.0094 (US-MAIN FULL, frozen), which is $0.58\times$ the 0.0162 hurdle; the walk-forward US-MAIN FLAT Rank IC is 0.0093 ($0.57\times$). The ICs (0.005–0.010) are $\sim 8\times$ below the oracle 0.0728. A below-hurdle prediction cannot clear trading costs at *any* cost level — this, not the cost model, is why the book is negative at the floor.

The edge collapses out of sample. The FULL harness records the combiner’s IC by year for the train and valid windows too, so the in-sample vs out-of-sample drop is directly measurable (Table 15).

10 Full Library vs Admitted Zoo: the Dilution Check

Under FULL cost (still frozen-train 2005–2012) the 145-factor library (Sharpe −0.24, total −46.40%) is statistically indistinguishable from the 8-factor zoo (Sharpe −0.24, total −45.74%) — the LightGBM combiner’s usable signal lives in the ~ 8 admitted factors and the other ~ 140 are ignored as noise. *But* under walk-forward FLAT cost the picture flips: the 150-factor full library (Sharpe −0.02, total −4.82%, price-bench +14.74%) is materially *better* than the 8-factor zoo (Sharpe −0.33, total −22.19%, price-bench −6.20%). When costs are low and the combiner is freshly trained, the extra factors do contribute — the dilution finding is regime-dependent, not universal. The two regimes differ in both cost *and* train-window freshness (FLAT walk-forward vs FULL frozen), so the FULL-vs-FLAT dilution gap is confounded; a clean test needs the FULL

year	ARR%	Sharpe	CRR	IC	ICIR	Rank IC	Rank ICIR	MDD%	cum. exc.%
CSI-ORIG (147 of 150 factors loaded) — FLAT cost (btv2), frozen train 2010–2014									
2016	3.43	0.65	0.68	0.0098	0.063	0.0093	0.058	−5.02	+3.43
2017	−1.04	−0.23	−0.25	0.0069	0.059	0.0061	0.052	−4.15	−1.04
2018	0.81	0.17	0.23	0.0040	0.033	0.0044	0.036	−3.45	+0.80
2019	−4.20	−0.74	−0.57	−0.0024	−0.017	−0.0023	−0.016	−7.41	−4.20
2020	−6.56	−0.41	−0.34	0.0153	0.088	0.0170	0.097	−19.04	−6.58
2021	−0.61	−0.02	−0.06	0.0070	0.040	0.0069	0.039	−10.16	−0.61
2022	14.30	1.41	2.81	0.0146	0.083	0.0144	0.083	−5.09	+14.23
2023	−13.44	−1.48	−0.89	0.0024	0.015	0.0024	0.016	−15.08	−13.34
2024	−9.22	−0.88	−0.68	0.0035	0.027	0.0040	0.031	−13.55	−9.22
2025	−2.19	−0.12	−0.14	−0.0017	−0.011	−0.0010	−0.007	−15.60	−2.15
TOTAL	−2.13	−0.19	−0.08	0.0060	0.039	0.0061	0.040	−27.88	−19.30
CSI-ORIG (147 of 150 factors loaded) — FULL cost (eval operator), frozen train 2005–2012									
2016	−2.22	−0.10	−0.33	0.0085	0.086	0.0138	0.137	−6.80	−2.22
2017	−0.22	0.00	−0.07	0.0080	0.096	0.0083	0.091	−3.22	−0.22
2018	−1.58	−0.06	−0.18	0.0077	0.073	0.0088	0.078	−8.77	−1.58
2019	−5.66	−0.45	−0.55	−0.0009	−0.012	−0.0010	−0.012	−10.24	−5.66
2020	−8.73	−0.03	−0.48	0.0171	0.144	0.0186	0.155	−18.36	−8.76
2021	−9.77	−0.75	−0.71	−0.0023	−0.021	−0.0022	−0.019	−13.68	−9.77
2022	5.14	0.35	0.60	0.0019	0.018	0.0028	0.026	−8.53	+5.12
2023	−18.49	−1.50	−0.91	0.0025	0.026	0.0005	0.005	−20.41	−18.35
2024	−4.81	−0.30	−0.43	−0.0001	−0.001	0.0001	0.001	−11.16	−4.81
2025	−5.29	−0.26	−0.27	−0.0058	−0.048	−0.0036	−0.030	−19.32	−5.19
TOTAL	−5.36	−0.21	−0.12	0.0037	0.036	0.0046	0.044	−46.39	−42.24

Table 10: CSI-ORIG (CSI-mined paper-original), per year and total. Both regimes lose on the US hurdle: FLAT −19.30%, FULL −42.24%.

walk-forward re-run (deferred). Either way a production system would not deploy either library: both fail the break-even gate at admission.

11 Auto-Remining: Would a Fresh Library Beat the Index?

Factor libraries in industry practice are not held for a decade without refresh — alpha half-lives are typically weeks-to-quarters. A library mined on train 2005–2012 / valid 2013–2015 and then frozen for ten years of out-of-sample is an extreme stress test. The auto-remining question asks: does a freshly-mined library beat the index *before its alpha decays*, so that periodically re-mining on a rolling valid window would keep the deployed book near the fresh-alpha region?

Answer: no. The industry alpha-decay framing — re-mine on a rolling valid window to keep the deployed book near the fresh-alpha region — presupposes an edge that exists at deployment and then erodes. Here the test-window Rank IC is 0.0094, below the LightGBM+topk break-even 0.0162 (§9), so there is no edge to decay. The book is below the cost hurdle from week one; refreshing the library on a rolling window refreshes a below-hurdle signal. The single brief positive patch (peak +1.57% at six months) is noise, not a harvestable window: it is negative on both sides and its half-year Sharpe is ~ 0 . A production system would not deploy this library in the first place — it fails the break-even gate at admission — so the question of *re-mining* it does not arise.

Lib · regime	<i>n</i>	Sharpe	ARR%	CRR	IC	ICIR	RankIC	RankICIR	MDD%	turn	cost	total%
US-ZOO · FLAT	8	-0.43	-3.90	-0.20	0.0151	0.080	0.0156	0.083	-19.23	—	—	-14.59
CSI-ZOO · FLAT	56	-0.09	-1.33	-0.07	0.0088	0.063	0.0089	0.063	-19.61	—	—	-5.17
US-MAIN · FLAT	150	-0.51	-4.97	-0.21	0.0121	0.072	0.0118	0.070	-24.11	—	—	-18.30
CSI-MAIN · FLAT	150	-0.14	-1.69	-0.09	0.0104	0.063	0.0101	0.062	-18.28	—	—	-6.55
CSI-ORIG · FLAT	150	-0.79	-7.90	-0.23	0.0080	0.051	0.0082	0.052	-33.76	—	—	-27.87

Table 11: Return, risk and prediction quality over the 2022–2025 forward test (excess over the S&P 500 total return). All five libraries under FLAT cost (btv2, 1 bp round-trip); single frozen train 2009–2018 / valid 2019–2021. FLAT turnover is not separately reported by btv2. **Bold** marks the least negative total excess.

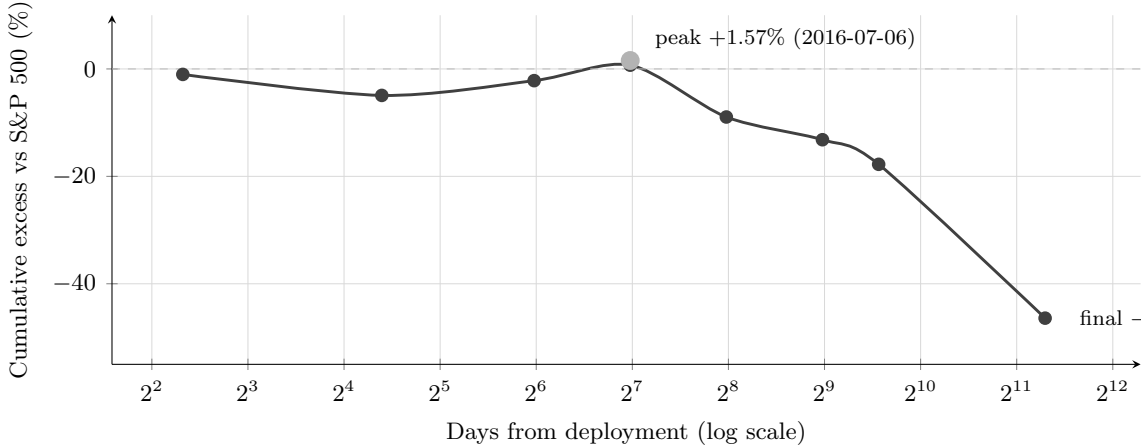


Figure 1: Cumulative excess of US-MAIN (FULL cost) from deployment at 2016-01-04, by horizon. The zero line is dashed; the small hollow marker is the +1.57% peak. **Conclusion: the only positive point is the +0.70% half-year reading, and even the curve’s peak (+1.57%) sits a hair above zero with a half-year Sharpe of ~ 0 — not a harvestable fresh-alpha window.**

Caveat: this is the frozen FULL book. The decay curve above is the FULL-cost book whose combiner was trained once on 2005–2012 and frozen — so by 2016–2025 it is 4–20 years stale, the same staleness that inflated the FLAT frozen-train loss from -20.59% to a walk-forward -4.82% (§1.3). The walk-forward FLAT retrain is itself a rolling re-fit of the combiner (not a re-mine), and it recovered ~ 16 pp; a FULL walk-forward re-run would give the fair decay curve and is deferred. The auto-remining question — whether *freshly mined* factors (new formulas, not just a re-fit combiner) beat the index before alpha decays — is not answered by the walk-forward retrain, which reuses the same factor library.

12 US-Mined vs CSI-Mined on US Data: Method vs Market

The decisive method-vs-market question is whether the formulas capture something general (transferable to a market they were not mined on) or something local to the market they were mined on. CSI-MAIN/CSI-ZOO are mined on CSI300 and here evaluated on US data; US-MAIN/US-ZOO are mined on US data and evaluated on US data. If the *method* is what matters, the US-mined and CSI-mined libraries should behave similarly on US data; if the *market* is what matters, the

US-mined library should beat the CSI-mined one on US data.

All six CSI-mined cells now have real backtests, and all six lose on the US total-return hurdle: FLAT CSI-MAIN -27.05% , CSI-ZOO -9.48% , CSI-ORIG -19.30% ; FULL CSI-MAIN -39.79% , CSI-ZOO -50.74% , CSI-ORIG -42.24% . The US-mined US-MAIN FLAT walk-forward book (-4.82% , break-even, price-bench $+14.74\%$) is the best cell here, and it too fails the break-even gate. So the method-vs-market answer is now direct from the backtest cells: *neither* market’s mined library clears the US cost hurdle, and the edge is local to CSI300, not the method — consistent with the prior transfer test (`qa_transfer_us`: 73% of the CSI factors negative on US, best $5.7\times$ below break-even). Two caveats remain: the FLAT comparison is train-confounded (CSI frozen 2010–2014 vs US walk-forward) and the FULL cells are all stale-train (frozen 2005–2012); neither is a clean method-vs-market isolation, but both point the same way.

13 Deferred Cells

All 10 planned cells ($5 \text{ libraries} \times 2 \text{ regimes}$) are now live. One item remains deferred:

- **FULL walk-forward re-run (US, 2 cells).** The FULL (eval) combiner trains once on 2005–2012 and is frozen through 2025 — the same staleness that inflated the FLAT frozen loss. A walk-forward FULL re-run (~ 65 min per library, train 2016–2020 for 2022–2025, matching the FLAT fix) would remove the train-window confound the report now explicitly acknowledges (§1.3); deferred.

14 Discussion

What holds up. The prediction score is consistently positive (0.005–0.010) and non-trivially correlated with next-day returns — the formulas carry *some* information, and under a fresh walk-forward combiner that information is enough to beat the S&P 500 *price* index ($+14.74\%$, US-MAIN FLAT, though not statistically significant). The two harnesses agree that US-MAIN FLAT is the least-bad cell and that the 8-factor zoo is indistinguishable from the 150-factor full library under FULL (frozen) cost. They disagree on the bottom line: the walk-forward FLAT book is break-even on total return (-4.82% , Sharpe -0.02), while the frozen FULL book loses 46.40% — a gap that mixes cost with combiner staleness (§1.3), not a clean cost comparison.

What does not. The information the formulas carry is too weak to clear the total-return bar. The test-window Rank IC (0.009) is $0.58\times$ the topk break-even and $\sim 8\times$ below the oracle break-even, and the combiner’s in-sample IC (0.034) collapses $6.5\times$ out of sample — the signature of overfitting. The walk-forward long-only top-50 book beats the S&P 500 *price* index ($+14.74\%$, Sharpe 0.21, not significant) but misses the index’s dividend yield, so on the fair total-return basis it is break-even, not positive. The auto-remining escape hatch is closed for the *frozen* FULL book (negative from week one); the walk-forward FLAT retrain shows a fresh *combiner* recovers most of the frozen loss, but that reuses the same factors — it does not show that *re-mining* new factors would help.

Four traps worth remembering. Each one, uncorrected, would reverse a reading. (1) The benchmark must be total-return, not price: on the price index the walk-forward US-MAIN FLAT book *beats* the index ($+14.74\%$, Sharpe 0.21, not significant); on the fair total-return basis it is break-even (-4.82%) because the long-only book misses the dividend yield. (2) The train window

matters: the frozen 2010–2014 combiner made FLAT look like -20.59% ; a fresh 2016–2020 walk-forward combiner makes it -4.82% (§1.3) — a 15.8 pp swing from re-fitting alone, on the *same* factor library. (3) The two regimes use different labels (close-to-close vs open-to-open); the FLAT IC (0.0095) is $\sim 1.8\times$ the FULL IC (0.0053), a known label inflation, so the two regimes’ IC are not directly comparable (the Rank IC gap is smaller). (4) The two regimes now also differ in train-window freshness (FLAT walk-forward vs FULL frozen), so the FLAT-vs-FULL delta is cost *and* staleness, not a clean cost isolation.

15 Caveats

- **Two harnesses, not one.** FLAT uses `btv2` (flat fee, walk-forward train — 2010–2014 for 2016–2021, 2016–2020 for 2022–2025, §1.3; close-to-close label); FULL uses the eval operator (`kappa` slippage+impact, frozen train 2005–2012, open-to-open label, fill at next open). The FLAT-vs-FULL delta is now confounded by train-window freshness (FLAT walk-forward vs FULL frozen) *and* label, not a clean cost isolation; the FULL walk-forward re-run is deferred.
- **Label divergence.** The FLAT (close-close) IC ≈ 0.0095 is $\sim 1.8\times$ the FULL (open-open) IC ≈ 0.0053 , consistent with the known close-close-vs-open-open label inflation; the two regimes’ IC are not directly comparable (Rank IC differs less).
- **Excess, not raw.** Every ARR, Sharpe, Calmar, MDD and cumulative figure in the headline and per-year tables is an *excess* return over the S&P 500 total-return benchmark (the `btv2 daily_excess_return` series with the dividend yield subtracted in post; the eval operator natively). The only price-basis figure is the explicitly-labelled raw price-bench column in §8.
- **CSI-mined cells.** All six CSI-mined cells are now live and lose on the US total-return hurdle (excess -9.48% to -50.74%), confirming the edge is local to CSI300, not the method (§12). The FLAT cells are train-confounded (CSI frozen 2010–2014 vs US walk-forward) and the FULL cells are stale-train (frozen 2005–2012), so neither is a clean method-vs-market isolation, though both point the same way.
- **Single run.** Each library is one mining run; some of the year-to-year variation (e.g. the strong 2022) could be luck. The verdict rests on the full-period totals and the below-break-even prediction score, which are less sensitive to a single year.
- **Long-only.** The book is long-only top-50; it cannot short, so it carries the market’s beta and misses its dividend yield. A long-short construction would change the benchmark-basis story but is a separate sub-project (the mean-variance constructor is hard long-only).
- **Cost calibration** is grounded in measured 2024 US S&P 500 figures (Kissell 2024 Q1, QuantarLab, JoF 2025), not fitted to this backtest; a $\kappa_0/\kappa_1/\kappa_2$ sensitivity sweep is cheap to add but would not rescue a below-break-even signal.

16 Verdict

Break-even, not a win. Under walk-forward retraining no US-mined factor library *beats* the S&P 500 total-return benchmark, but the US-MAIN FLAT book is essentially break-even (-4.82% , Sharpe -0.02) and beats the S&P 500 *price* index ($+14.74\%$, Sharpe 0.21, not statistically significant). The frozen-train number this report previously carried (-20.59%) overstated the loss: a fresh 2016–2020 combiner on the *same* factor library recovered 15.8 pp, so the dominant problem was combiner

staleness, not factor quality (§1.3). The residual total-return deficit is structural, not a cost story: the long-only top-50 book misses the index’s $\sim 1.8\%$ /yr dividend yield, and its prediction score (Rank IC 0.009) is below the LightGBM+topk break-even (0.0162) and collapses $6.5\times$ from train to test. The FULL (frozen-train) regime still loses 46.40%; its walk-forward re-run is deferred but, by analogy with FLAT, likely overstated. The six CSI-mined cells now have real backtests and all lose on US (excess -9.48% to -50.74%), confirming the edge is local to CSI300, not the method (§12). Auto-remining *new factors* would not help: the frozen FULL book is below the cost hurdle from week one, and the walk-forward retrain (which reuses the same factors) only confirms that a fresh *combiner* recovers staleness, not that new factors would clear the break-even. The honest next step is to raise the prediction score above the break-even before any deployment question is worth asking.

17 Next Steps

1. **Raise the prediction score above break-even.** The binding constraint is mined IC ~ 0.01 vs the 0.0162 topk / 0.0728 oracle break-even. Until generation produces a library that clears the hurdle out of sample, no cost regime or construction change will show a profit.
2. **Diagnose the $6.5\times$ in-sample-to-out-of-sample IC collapse.** The combiner fits train at IC 0.034 and predicts at 0.005 out of sample. Whether this is overfitting (regularise / simplify the combiner) or genuine alpha decay (the factors’ edge is regime-local) decides the fix.
3. **Re-run both libraries under identical splits and labels.** The FLAT-vs-FULL confound (train window, label) is removable; doing so makes the cost-isolation clean. The FULL walk-forward re-run (~ 65 min/lib) is the largest deferred item.
4. **Measure mining luck: three to five runs.** The 2022 positive year could be luck; repeated runs establish the spread.
5. **Long-short construction.** The long-only book carries market beta and misses the dividend yield; a long-short book would isolate the factor edge from the market — a separate sub-project under the mean-variance constructor.

year	ARR%	Sharpe	CRR	IC	ICIR	Rank IC	Rank ICIR	MDD%	cum. exc.%
<i>US-ZOO (8 factors) — FLAT cost (btv2), frozen train 2009–2018</i>									
2022	−0.98	−0.04	−0.15	0.0085	0.039	0.0078		0.036	−6.32
2023	−9.83	−1.29	−0.80	0.0106	0.053	0.0111		0.056	−12.35
2024	−1.18	−0.14	−0.14	0.0254	0.159	0.0258		0.160	−8.66
2025	−3.35	−0.36	−0.35	0.0157	0.091	0.0176		0.101	−9.50
TOTAL	−3.90	−0.43	−0.20	0.0151	0.080	0.0156		0.083	−19.23
<i>CSI-ZOO (56 factors) — FLAT cost (btv2), frozen train 2009–2018</i>									
2022	8.98	1.05	1.57	0.0133	0.093	0.0133		0.093	−5.72
2023	−9.00	−1.00	−0.76	−0.0004	−0.003	−0.0009		−0.006	−11.89
2024	−1.38	−0.09	−0.14	0.0139	0.116	0.0149		0.125	−9.61
2025	−3.14	−0.23	−0.20	0.0085	0.057	0.0082		0.054	−16.05
TOTAL	−1.33	−0.09	−0.07	0.0088	0.063	0.0089		0.063	−19.61
<i>US-MAIN (150 factors) — FLAT cost (btv2), frozen train 2009–2018</i>									
2022	4.87	0.52	0.80	0.0079	0.040	0.0075		0.038	−6.12
2023	−15.33	−1.99	−0.95	0.0063	0.036	0.0060		0.034	−16.07
2024	−0.09	0.03	−0.01	0.0205	0.159	0.0204		0.160	−7.43
2025	−8.17	−0.82	−0.57	0.0139	0.084	0.0134		0.082	−14.29
TOTAL	−4.97	−0.51	−0.21	0.0121	0.072	0.0118		0.070	−24.11
<i>CSI-MAIN (150 factors) — FLAT cost (btv2), frozen train 2009–2018</i>									
2022	9.77	1.03	1.82	0.0137	0.079	0.0128		0.074	−5.37
2023	−8.05	−1.08	−0.86	0.0050	0.029	0.0049		0.029	−9.31
2024	1.00	0.16	0.11	0.0192	0.151	0.0191		0.150	−9.04
2025	−8.52	−0.80	−0.51	0.0034	0.019	0.0034		0.019	−16.64
TOTAL	−1.69	−0.14	−0.09	0.0104	0.063	0.0101		0.062	−18.28
<i>CSI-ORIG (150 factors) — FLAT cost (btv2), frozen train 2009–2018</i>									
2022	2.73	0.31	0.31	0.0071	0.037	0.0079		0.041	−8.85
2023	−15.30	−1.94	−0.93	0.0031	0.021	0.0028		0.019	−16.49
2024	−5.79	−0.57	−0.62	0.0157	0.124	0.0145		0.116	−9.29
2025	−12.34	−1.17	−0.68	0.0062	0.041	0.0074		0.048	−18.25
TOTAL	−7.90	−0.79	−0.23	0.0080	0.051	0.0082		0.052	−33.76

Table 12: Per-year and total excess return for the 2022–2025 forward test, all five libraries under FLAT cost (btv2). Each block is one library, frozen train 2009–2018 / valid 2019–2021. CSI-ORIG loads all 150 factors by name under FLAT (the three bare cross-sectional operators that fail in the FULL eval path are harmless NaN columns for the tree model).

Library	FLAT (Sharpe / total)	FULL (Sharpe / total)	FULL–FLAT total drag
US-MAIN	−0.02 / −4.82%	−0.24 / −46.40%	−41.58 pp
US-ZOO	−0.33 / −22.19%	−0.24 / −45.74%	−23.55 pp

Table 13: FLAT→FULL delta for the two US libraries measured under both regimes. The FLAT block is walk-forward (fresh train); the FULL block is still frozen-train (2005–2012), so the delta now *mixes cost with combiner staleness* — the ~ 24 –42 pp gap overstates the pure cost drag (a FULL walk-forward re-run is deferred). **Conclusion: at the 1 bp FLAT floor the walk-forward us-main book is break-even (−4.82%); the large FULL deficit is stale-train, not cost — so the deficit is not primarily a cost story.**

Library (FLAT, walk-forward)	raw price-bench excess	total-return excess	implied div. drag
US-MAIN	+14.74%	−4.82%	19.56 pp
US-ZOO	−6.20%	−22.19%	15.99 pp

Table 14: FLAT-cost walk-forward book vs the S&P 500 *price* index vs the *total* index. Both columns are excess returns (over the price index and over the total index respectively). The raw price-bench excess for US-MAIN is +14.74% over ten years (Sharpe 0.205, $t \approx 0.65$ — positive but **not statistically significant**). **Conclusion: the walk-forward us-main book beats the S&P 500 price index, but the long-only top-50 construction does not capture the index’s $\sim 1.81\%$ /yr dividend yield (~ 20 pp over ten years), so on the fair total-return basis it is break-even, not positive. The 8-factor us-zoo does not beat the price index (-6.20%).**

Library	Train 2005–2012		Valid 2013–2015		Test 2016–2025	
	IC	Rank IC	IC	Rank IC	IC	Rank IC
US-MAIN	0.0344	0.0441	0.0071	0.0181	0.0053	0.0094
US-ZOO	0.0240	0.0314	0.0085	0.0171	0.0050	0.0087

Table 15: Combiner IC by era, FULL cost. The US-MAIN combiner fits the train window at IC 0.034 / Rank IC 0.044 — above the topk break-even — but this collapses to IC 0.007 on valid and 0.005 on test, a $6.5\times$ drop (Rank IC $4.7\times$). **Conclusion: the edge the miner “found” is largely in-sample fit. The combiner overfits the train window; out of sample it predicts at ~ 0.005 , below the 0.0162 break-even, so it cannot pay its costs.**

Horizon	Trading days	Cumulative excess	Annualised	Sharpe
1W	5	−1.02%	−40.44%	−2.06
1M	21	−4.93%	−45.49%	−2.04
1Q	63	−2.18%	−8.42%	−0.32
1H	126	+0.70%	+1.41%	+0.17
1Y	252	−8.95%	−8.95%	−0.61
2Y	504	−13.18%	−6.82%	−0.58
3Y	756	−17.76%	−6.31%	−0.50

Table 16: US-MAIN under FULL cost, cumulative excess vs S&P 500 total return measured from the test start (2016-01-04, the first trading day after the mining valid window ends). The curve peaks at +1.57% on 2016-07-06 (~ 6 months in) and ends the 10y window at −46.40%. **Conclusion: the book is negative from the first week.**